

Multivariable Calculus

MATH 1920 Lecture Notes

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1 Vectors in Two and Three Dimensions

Question: What information specifies a line?

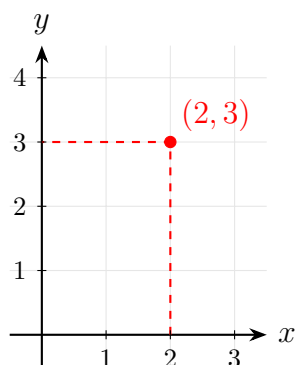
- 2 points
- A point and a direction vector
- Slope and y -intercept
- Intersection of 2 non-parallel planes

Comparison:

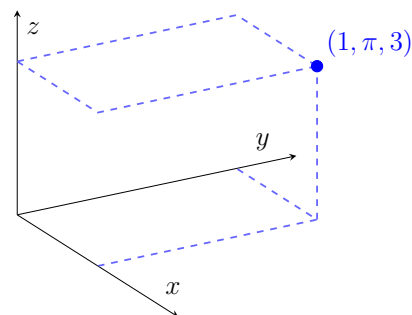
- **Single-variable calculus:** $f(x) \rightarrow$ rate of change of x
- **Multivariable calculus:** $f(x, y) \rightarrow$ functions of multiple variables

1.1 Coordinates

2-dimension (2D): $(2, 3)$



3-dimension (3D): $(1, \pi, 3)$



1.2 Vectors and Position Vectors

Definition 1.1: Equivalent Vectors

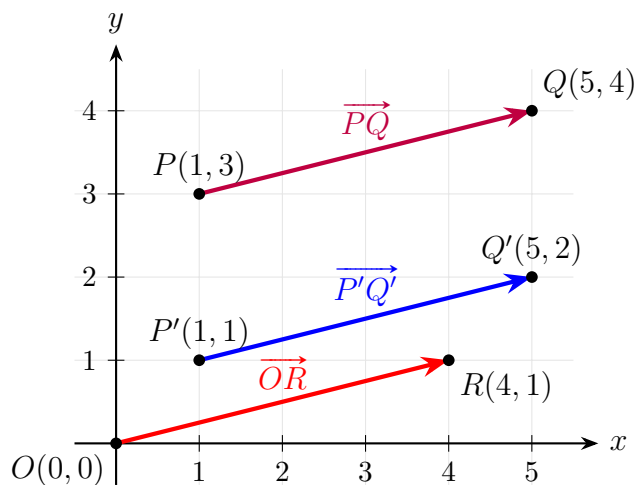
Two vectors are **equivalent** if one is a translation (pure shift without rotation) of the other.

Example 1.1. Given points $P = (1, 3)$, $Q = (5, 4)$ and $P' = (1, 1)$, $Q' = (5, 2)$. Let

$R = (4, 1)$ and $O = (0, 0)$. Then:

$$\overrightarrow{PQ} = \overrightarrow{P'Q'} = \overrightarrow{OR}$$

This could be indicated by the diagram below:



Definition 1.2: Position Vector

A vector whose initial point is at the origin $O(0, 0)$ is called a **position vector**. It is written using angle brackets:

$$\overrightarrow{OR} = \langle 4, 1 \rangle$$

Definition 1.3: Magnitude

The **length** or **magnitude** of a vector $\vec{v}$, denoted by $\|\vec{v}\|$, is the length of the line segment between its endpoints.

If $\vec{v} = \langle v_1, v_2 \rangle$, then its magnitude is given by:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$$

For example, using $\overrightarrow{PQ} = \langle 4, 1 \rangle$:

$$\|\overrightarrow{PQ}\| = \sqrt{4^2 + 1^2} = \sqrt{17}$$

1.3 Unit Vectors

A vector $\vec{v}$ is called a **unit vector** if its magnitude is 1, i.e., $\|\vec{v}\| = 1$.

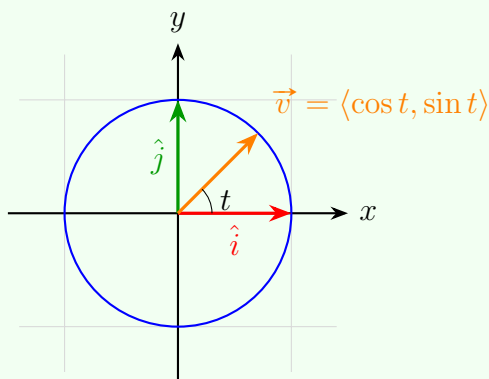
Problem 1.1: Describing Unit Vectors

Describe all unit vectors in $\mathbb{R}^2$.

Solution 1.1

A vector $\vec{v} = \langle x, y \rangle$ is a unit vector if and only if $x^2 + y^2 = 1$. Equivalently, using polar parameterization:

$$\vec{v}(t) = \langle \cos t, \sin t \rangle \quad \text{for } 0 \leq t < 2\pi$$



Standard Basis Vectors:

- **2D Basis Vectors:** $\hat{i} = \langle 1, 0 \rangle$, $\hat{j} = \langle 0, 1 \rangle$
- **3D Basis Vectors:** $\hat{i} = \langle 1, 0, 0 \rangle$, $\hat{j} = \langle 0, 1, 0 \rangle$, $\hat{k} = \langle 0, 0, 1 \rangle$

1.4 Vector Operations

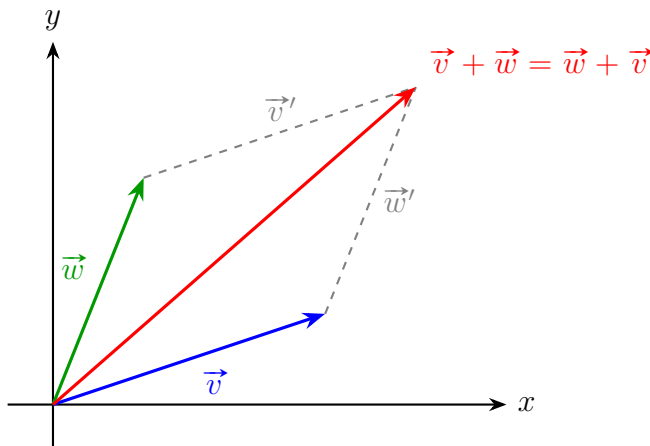
Addition and Subtraction:

$$\langle v_1, v_2 \rangle + \langle w_1, w_2 \rangle = \langle v_1 + w_1, v_2 + w_2 \rangle$$

$$\langle v_1, v_2 \rangle - \langle w_1, w_2 \rangle = \langle v_1 - w_1, v_2 - w_2 \rangle \quad (\text{Equivalent to } \vec{v} + (-1)\vec{w})$$

Scalar Multiplication: For any scalar $\lambda \in \mathbb{R}$ and vector $\vec{v} = \langle v_1, v_2 \rangle$:

$$\lambda \vec{v} = \langle \lambda v_1, \lambda v_2 \rangle$$



Important Note

Vectors are **not** designed to be divided! (Vector division is undefined).

1.5 Parallel Vectors

Definition 1.4: Parallel Vectors

Two non-zero vectors $\vec{v}$ and $\vec{w}$ ($\vec{v}, \vec{w} \neq \langle 0, 0 \rangle$) are *parallel* (denoted as $\vec{v} \parallel \vec{w}$) if there exists a non-zero scalar $\lambda \in \mathbb{R}$ such that:

$$\vec{w} = \lambda \vec{v}$$

Equivalent Geometric Definition: $\vec{v} \parallel \vec{w}$ if and only if both vectors define or lie on the *same line passing through the origin*.

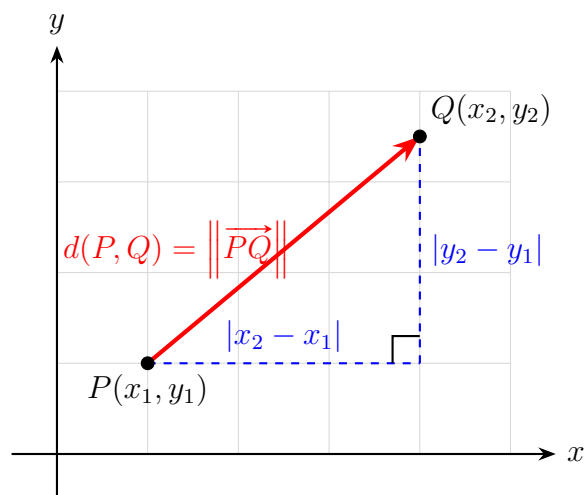
2 Discussion Week 1: Distance Formulae and Geometric Shapes

Definition 2.1: Distance Formula in $\mathbb{R}^2$

Let $(x, y) \in \mathbb{R}^2$ and points $P = (x_1, y_1)$, $Q = (x_2, y_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\| \overrightarrow{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This distance formula is a direct application of the Pythagorean theorem, where the distance $d(P, Q)$ forms the hypotenuse of a right-angled triangle with legs $|x_2 - x_1|$ and $|y_2 - y_1|$, as shown in the figure below.



The case in $\mathbb{R}^3$ follows similarly:

Definition 2.2: Distance Formula in $\mathbb{R}^3$

Let $(x, y, z) \in \mathbb{R}^3$ and points $P = (x_1, y_1, z_1)$, $Q = (x_2, y_2, z_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\| \overrightarrow{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

2.1 Equations of Geometric Shapes

Consider the equation $x^2 + y^2 = 4$.

In $\mathbb{R}^2$, this equation is equivalent to the set:

$$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$$

This represents a circle with radius $r = 2$ centered at the origin. The distance from the origin $(0, 0)$ to any point (x, y) on this circle is constant and equal to 2, which can be verified directly using the distance formula:

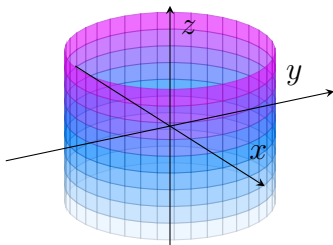
$$\sqrt{(x - 0)^2 + (y - 0)^2} = 2$$

Hence, the geometry of the circle is directly implied by the Pythagorean theorem.

Now consider the equation $x^2 + y^2 = 1$ in three-dimensional space, $\mathbb{R}^3$. This set is formally expressed as:

$$\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$$

Because z is completely unrestricted, the cross-section at any constant height $z = c$ forms the unit circle $x^2 + y^2 = 1$. Consequently, this equation describes a *right circular cylinder* extending infinitely along the z -axis, as shown.



2.2 Linear Equations across Dimensions

In $\mathbb{R}^2$, the general form of a linear equation is given by:

$$Ax + By + C = 0 \quad (A, B, C \in \mathbb{R}, \text{ with } A \text{ and } B \text{ not both zero})$$

This equation represents a *line* in two-dimensional space.

In $\mathbb{R}^3$, however, the exact same algebraic equation $Ax + By + C = 0$ represents a *plane* parallel to the z -axis (since z is a free variable), not a line.

To uniquely define a line in three-dimensional space, a single linear equation is insufficient; we require a system of *two* independent linear equations (representing the intersection of two non-parallel planes).

3 Vectors and Directions

Problem 3.1: Vector Directions

Let $\vec{v} = \langle 3, -4, 2 \rangle$. Find a unit vector in the same direction as $\vec{v}$.

Solution 3.1

To find a unit vector in the same direction as $\vec{v}$, we first compute its magnitude:

$$\|\vec{v}\| = \sqrt{3^2 + (-4)^2 + 2^2} = \sqrt{9 + 16 + 4} = \sqrt{29}$$

The unit vector in the same direction as $\vec{v}$ is then:

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\sqrt{29}} \langle 3, -4, 2 \rangle$$

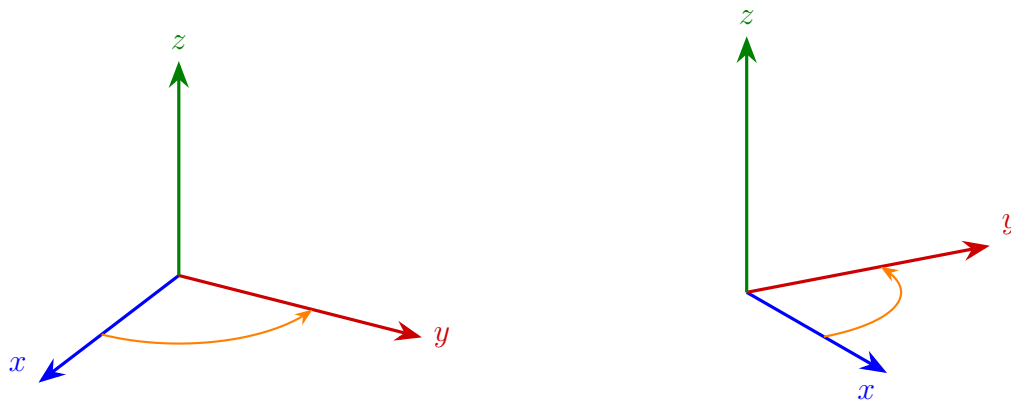
In general, for any non-zero vector $\vec{w}$,

$$\begin{aligned}\|\lambda \vec{w}\| &= \sqrt{(\lambda w_1)^2 + (\lambda w_2)^2 + (\lambda w_3)^2} \\ &= \sqrt{\lambda^2(w_1^2 + w_2^2 + w_3^2)} \\ &= |\lambda| \|\vec{w}\|\end{aligned}$$

Hence, if we want a unit vector in the same direction as $\vec{w}$, we can choose $\lambda = \frac{1}{\|\vec{w}\|}$, giving us:

$$\hat{w} = \frac{\vec{w}}{\|\vec{w}\|}$$

Right-Hand Rule The relative position of the x -, y -, and z -axes in $\mathbb{R}^3$ is determined by the *right-hand rule*. If you point your right thumb along the positive x -axis and your index finger along the positive y -axis, then your middle finger (perpendicular to both) will point along the positive z -axis.



Definition 3.1: Parameterizations

A parameterization is a function whose range is a given geometric object.

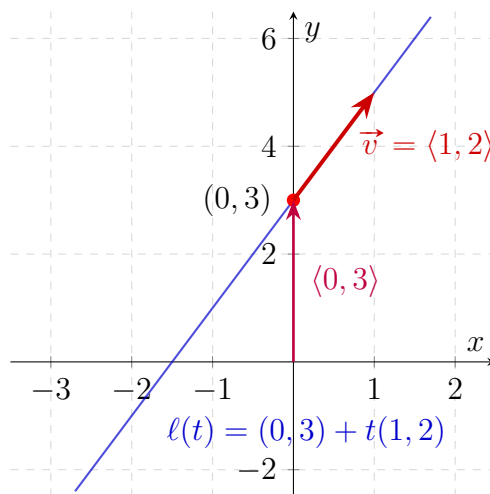
For example, a parameterization of the unit circle is given by:

$$f(t) = (\cos t, \sin t)$$

Example 3.1. To parameterize the line $y = 2x + 3$:

$$\begin{aligned}\ell(t) &= (t, 2t + 3) \\ &= \langle 0, 3 \rangle + \langle t, 2t \rangle\end{aligned}$$

The plot is then given:



To generalize, to parameterize a line in $\mathbb{R}^2$, we can first take a point on the line. Then, we

take some $\vec{v}$ in the direction of the line. Then, we can write the parameterization as:

$$\ell(t) = (p_1 + v_1t, p_2 + v_2t)$$

In $\mathbb{R}^3$, we can do the same thing.

Example 3.2. Parameterize the line passing through $P(1, 2, 0)$ and $Q(5, -1, 1)$.

$$\begin{aligned}\overrightarrow{PQ} &= \langle 5 - 1, -1 - 2, 1 - 0 \rangle \\ &= \langle 4, -3, 1 \rangle \\ \ell(t) &= (1, 2, 0) + t(4, -3, 1) \\ &= (1 + 4t, 2 - 3t, t)\end{aligned}$$

3.1 Dot Product

Definition 3.2: Dot Product

The dot product of two vectors $\vec{v} = \langle v_1, v_2 \rangle$ and $\vec{w} = \langle w_1, w_2 \rangle$ is defined as:

$$\vec{v} \cdot \vec{w} = v_1w_1 + v_2w_2$$

Example 3.3.

$$\begin{aligned}\vec{v} \cdot \vec{v} &= v_1^2 + v_2^2 = \|\vec{v}\|^2 \\ \vec{i} \cdot \vec{j} &= 1 \cdot 0 + 0 \cdot 1 = 0\end{aligned}$$

Problem 3.2: Dot Product and Angle

Find pairs of vectors with:

- Dot product > 0
- Dot product $= 0$
- Dot product < 0

Solution 3.2

- $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2 > 0$ for any non-zero vector $\vec{v}$.
- $\vec{i} \cdot \vec{j} = 0$ since they are orthogonal.
- $\langle -1, -1 \rangle \cdot \langle 1, 1 \rangle = -1 - 1 = -2 < 0$.

To conclude, the dot products of any two vectors corresponds to the following situations:

- Dot product = 0 if and only if the two vectors are orthogonal (perpendicular).
- Dot product > 0 if the angle between the two vectors is acute (less than 90 degrees).
- Dot product < 0 if the angle between the two vectors is obtuse (greater than 90 degrees).

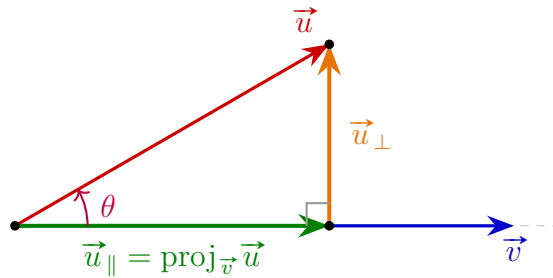
This is because $\vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos \theta$, where θ is the angle between the vectors. Hence,

$$\theta = \arccos \left(\frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \|\vec{w}\|} \right)$$

Definition 3.3: Orthogonal Projection

Write $\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$, where $\vec{u}_{\parallel} \parallel \vec{v}$ and $\vec{u}_{\perp} \perp \vec{u}_{\parallel}$. Then, $\vec{u}_{\parallel}$ is called the **orthogonal projection** of $\vec{u}$ onto $\vec{v}$.

Example 3.4. Below is an illustration of orthogonal projection:



By this, we could see that

$$\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$$

Then, we solve for $\vec{u}_{\parallel}$ and $\vec{u}_{\perp}$.

Taking the dot product with $\vec{v}$:

$$\begin{aligned}
 \vec{u} \cdot \vec{v} &= (\vec{u}_{\parallel} + \vec{u}_{\perp}) \cdot \vec{v} \\
 &= \vec{u}_{\parallel} \cdot \vec{v} + \vec{u}_{\perp} \cdot \vec{v} \\
 &= \vec{u}_{\parallel} \cdot \vec{v} + 0 \quad (\text{since } \vec{u}_{\perp} \perp \vec{v}) \\
 &= \vec{u}_{\parallel} \cdot \vec{v}
 \end{aligned}$$

Let $\vec{e}_v = \frac{1}{\|\vec{v}\|} \vec{v}$ be the unit vector in the direction of $\vec{v}$, and write $\vec{u}_{\parallel} = \lambda \vec{e}_v$. Then:

$$\begin{aligned}
 \vec{u} \cdot \vec{v} &= (\lambda \vec{e}_v) \cdot (\|\vec{v}\| \vec{e}_v) \\
 &= \lambda \|\vec{v}\| (\vec{e}_v \cdot \vec{e}_v) \\
 &= \lambda \|\vec{v}\| \|\vec{e}_v\|^2 \\
 &= \lambda \|\vec{v}\| \quad (\text{since } \|\vec{e}_v\| = 1)
 \end{aligned}$$

Solving for λ :

$$\lambda = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|}$$

Thus, the parallel projection vector $\vec{u}_{\parallel}$ and the orthogonal component $\vec{u}_{\perp}$ are given by:

$$\begin{aligned}
 \vec{u}_{\parallel} &= \left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|} \right) \vec{e}_v \\
 &= \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} \\
 &= \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} \\
 \vec{u}_{\perp} &= \vec{u} - \vec{u}_{\parallel} \\
 &= \vec{u} - \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v}
 \end{aligned}$$

Furthermore, the magnitude of the parallel projection is:

$$\|\vec{u}_{\parallel}\| = \|\vec{u}\| |\cos \theta|$$

Problem 3.3: Vector Projection

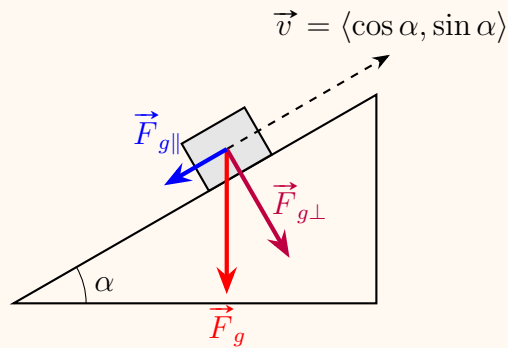
Find the projection of $\vec{u} = \langle 3, 2, 2 \rangle$ onto $\vec{v} = \langle 1, -2, 0 \rangle$

Solution 3.3

$$\begin{aligned}\vec{u}_{\parallel} &= \frac{3 \cdot 1 + 2 \cdot (-1) + 2 \cdot 0}{1^2 + (-1)^2 + 0^2} \langle 1, -1, 0 \rangle \\ &= \frac{1}{2} \langle 1, -1, 0 \rangle \\ &= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle\end{aligned}$$

Problem 3.4: Acceleration on an Inclined Plane

Consider a block sliding down an inclined plane with an angle of inclination α .



Solution 3.4

To find the magnitude of acceleration down the incline, we project the gravitational force $\vec{F}_g$ along the incline direction:

$$\begin{aligned}\text{acceleration} &= \left\| \vec{F}_{g\parallel} \right\| \\ &= \left\| \vec{F}_g \right\| |\cos \theta| \\ &= \left\| \vec{F}_g \right\| \sin \alpha = 9.8 \sin \alpha \quad (\text{m/s}^2)\end{aligned}$$

Problem 3.5: Vector Projection

Find the projection of $\vec{u} = \langle 3, 2, 2 \rangle$ onto $\vec{v} = \langle 1, -1, 0 \rangle$.

Solution 3.5

Using the vector projection formula:

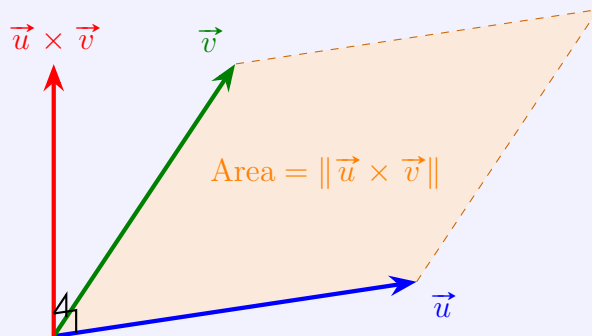
$$\begin{aligned}\vec{u}_{\parallel} &= \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} \\ &= \frac{3(1) + 2(-1) + 2(0)}{1^2 + (-1)^2 + 0^2} \langle 1, -1, 0 \rangle \\ &= \frac{3 - 2 + 0}{1 + 1 + 0} \langle 1, -1, 0 \rangle \\ &= \frac{1}{2} \langle 1, -1, 0 \rangle \\ &= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle\end{aligned}$$

3.2 Determinants and Cross Products

Definition 3.4: Vector Cross Product

Let $\vec{u}, \vec{v} \in \mathbb{R}^3$. The **cross product** $\vec{u} \times \vec{v}$ is a vector in $\mathbb{R}^3$ satisfying the following three geometric properties:

1. **Orthogonality:** $\vec{u} \times \vec{v} \perp \vec{u}$ and $\vec{u} \times \vec{v} \perp \vec{v}$.
2. **Magnitude:** $\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin \theta$, which equals the area of the parallelogram formed by $\vec{u}$ and $\vec{v}$.
3. **Orientation:** The ordered triple $(\vec{u}, \vec{v}, \vec{u} \times \vec{v})$ is right-handed (positively oriented).



From Property (2), we have for the standard basis vectors:

$$\|\vec{i} \times \vec{i}\| = \|\vec{i}\| \|\vec{i}\| \sin 0^\circ = 0 \implies \vec{i} \times \vec{i} = \vec{0}$$

And by the Right-Hand Rule:

$$\vec{i} \times \vec{j} = \vec{k}$$

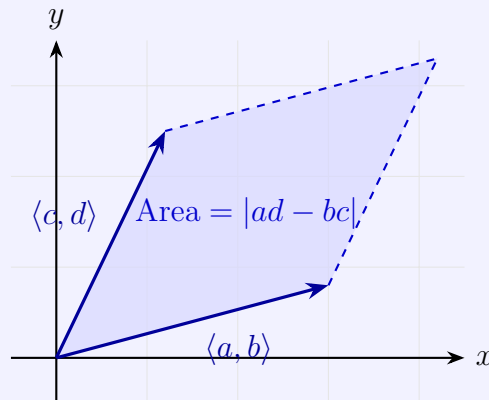
Definition 3.5: Determinants

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is defined as:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

This is denoted as $\det(A)$ or $|A|$ for a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

For a 2×2 matrix, the determinant represents the area of the parallelogram spanned by $\langle a, c \rangle$ and $\langle b, d \rangle$ (the column vectors) or $\langle a, b \rangle$ and $\langle c, d \rangle$.

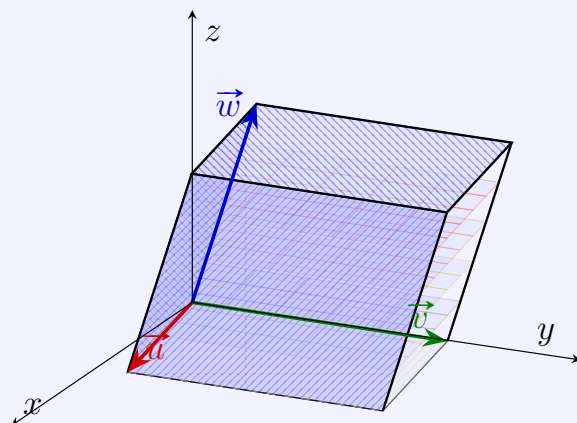


For a 3×3 matrix $B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, the determinant is defined as:

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= aei + bfg + cdh - ceg - bdi - afh$$

$\det \begin{pmatrix} \vec{u} \\ \vec{v} \\ \vec{w} \end{pmatrix}$ represents the volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$, and $\vec{w}$.



The cross product of two vectors $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$ can be computed using the determinant of a 3×3 matrix:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \vec{i} \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \vec{j} \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \vec{k} \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

Problem 3.6: Cross Product

Let $\vec{v} = \langle 3, 2, -1 \rangle$, $\vec{w} = \langle 4, 1, -3 \rangle$. Find $\vec{v} \times \vec{w}$.

Solution 3.6

$$\begin{aligned}\vec{v} \times \vec{w} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 2 & -1 \\ 4 & 1 & -3 \end{vmatrix} \\ &= \vec{i} \begin{vmatrix} 2 & -1 \\ 1 & -3 \end{vmatrix} - \vec{j} \begin{vmatrix} 3 & -1 \\ 4 & -3 \end{vmatrix} + \vec{k} \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} \\ &= \vec{i}(2(-3) - (-1)(1)) - \vec{j}(3(-3) - (-1)(4)) + \vec{k}(3(1) - 2(4)) \\ &= \vec{i}(-6 + 1) - \vec{j}(-9 + 4) + \vec{k}(3 - 8) \\ &= \vec{i}(-5) - \vec{j}(-5) + \vec{k}(-5) \\ &= -5\vec{i} + 5\vec{j} - 5\vec{k}\end{aligned}$$

3.2.1 Properties of the Cross Product

- $\vec{v} \times \vec{w} \perp \vec{v}$ and $\vec{v} \times \vec{w} \perp \vec{w}$. This implies that $\vec{v} \cdot (\vec{v} \times \vec{w}) = 0$ and $\vec{w} \cdot (\vec{v} \times \vec{w}) = 0$.
- $\{\vec{v}, \vec{w}, \vec{v} \times \vec{w}\}$ forms a right-handed system.
- $\|\vec{v} \times \vec{w}\|$ represents the area of the parallelogram spanned by $\vec{v}$ and $\vec{w}$.

3.2.2 Algebraic Rules of the Cross Product

- Anti-commutativity: $\vec{v} \times \vec{w} = -(\vec{w} \times \vec{v})$
- Parallel vectors: $\vec{v} \times \vec{w} = \vec{0} \iff \vec{v} = \lambda\vec{w} \text{ or } \vec{w} = \vec{0}$.
- Scalar multiplication: $\lambda(\vec{v} \times \vec{w}) = (\lambda\vec{v}) \times \vec{w} = \vec{v} \times (\lambda\vec{w})$
- Distributivity: $(\vec{u} + \vec{v}) \times \vec{w} = \vec{u} \times \vec{w} + \vec{v} \times \vec{w}$
- Standard basis relations:

$$\vec{i} \times \vec{j} = \vec{k}, \quad \vec{j} \times \vec{k} = \vec{i}, \quad \vec{k} \times \vec{i} = \vec{j}$$

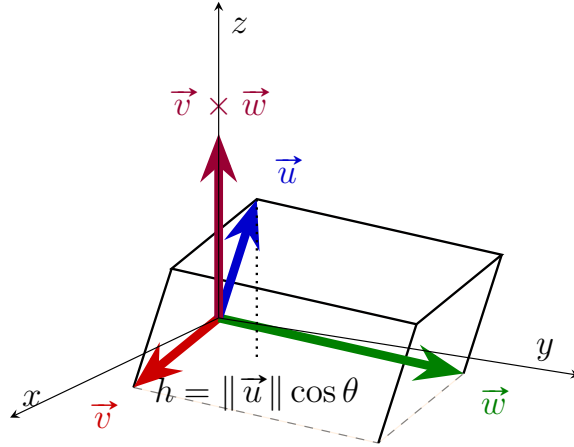
Hence, we may expand the cross product of two vectors $\vec{v} = \langle v_1, v_2, v_3 \rangle$ and $\vec{w} = \langle w_1, w_2, w_3 \rangle$ component-wise:

$$\begin{aligned}
\vec{v} \times \vec{w} &= (v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k}) \times (w_1 \vec{i} + w_2 \vec{j} + w_3 \vec{k}) \\
&= v_1 w_1 (\vec{i} \times \vec{i}) + v_1 w_2 (\vec{i} \times \vec{j}) + v_1 w_3 (\vec{i} \times \vec{k}) \\
&\quad + v_2 w_1 (\vec{j} \times \vec{i}) + v_2 w_2 (\vec{j} \times \vec{j}) + v_2 w_3 (\vec{j} \times \vec{k}) \\
&\quad + v_3 w_1 (\vec{k} \times \vec{i}) + v_3 w_2 (\vec{k} \times \vec{j}) + v_3 w_3 (\vec{k} \times \vec{k}) \\
&= v_1 w_2 \vec{k} - v_1 w_3 \vec{j} - v_2 w_1 \vec{k} + v_2 w_3 \vec{i} + v_3 w_1 \vec{j} - v_3 w_2 \vec{i} \\
&= (v_2 w_3 - v_3 w_2) \vec{i} + (v_3 w_1 - v_1 w_3) \vec{j} + (v_1 w_2 - v_2 w_1) \vec{k}
\end{aligned}$$

3.3 Scalar Triple Product

Given $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$, the quantity $\vec{u} \cdot (\vec{v} \times \vec{w})$ is defined as the **scalar triple product**.

The absolute value $|\vec{u} \cdot (\vec{v} \times \vec{w})|$ gives the geometric volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$, and $\vec{w}$:



The sign of $\vec{u} \cdot (\vec{v} \times \vec{w})$ indicates the orientation of the vector basis.

$\vec{u} \cdot (\vec{v} \times \vec{w}) > 0$ if the ordered triple $(\vec{u}, \vec{v}, \vec{w})$ forms a **right-handed system** (i.e., $\vec{u}$ points to the same side of the plane spanned by $\vec{v}$ and $\vec{w}$ as $\vec{v} \times \vec{w}$).

4 Discussion Week 2: Quadric Surfaces and Traces

Definition 4.1: Quadric Surface

A quadric surface is defined by a quadratic equation in three variables x , y , and z :

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + ax + by + cz + d = 0$$

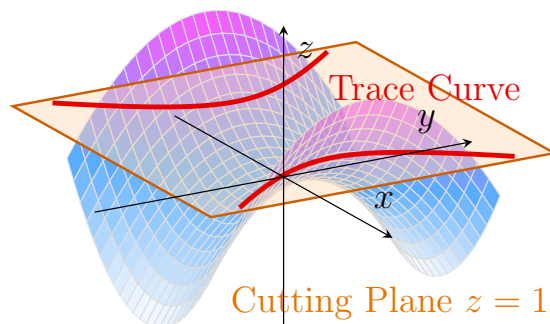
, where $A, B, C, D, E, F, a, b, c, d$ are constants, and at least one of A, B, C, D, E, F is non-zero.

Definition 4.2: Trace

A trace is the intersection of a quadric surface with a plane, usually by taking slices parallel to the coordinate planes.

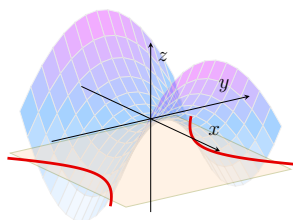
Note that when our surface can be written by $z = f(x, y)$, the traces form level sets of f .

Example 4.1. Consider the quadratic equation $z = x^2 - y^2$. Now take the traces in $z = -1$, $z = 0$, and $z = 1$. Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. In general, the traces in $x = k$ when k is a constant are hyperbolas with equations $z = k^2 - y^2$.

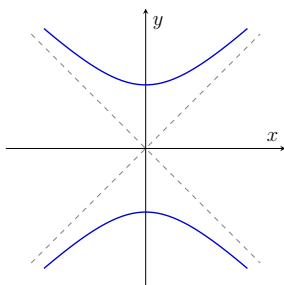


Now take the traces in $z = -1$, $z = 0$, and $z = 1$.

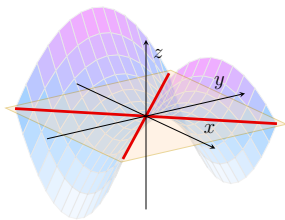
Trace at $z = -1$



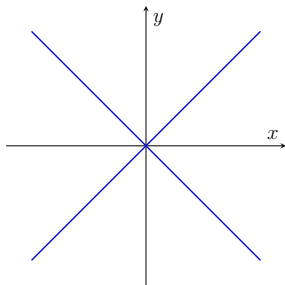
$$y^2 - x^2 = 1$$



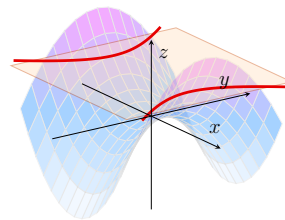
Trace at $z = 0$



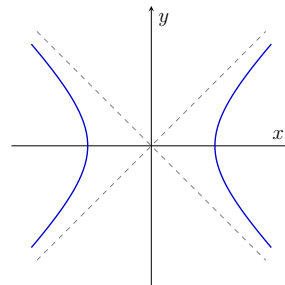
$$y = \pm x$$



Trace at $z = 1$

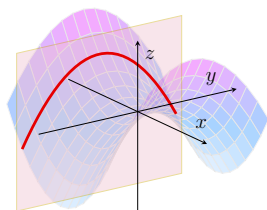


$$x^2 - y^2 = 1$$

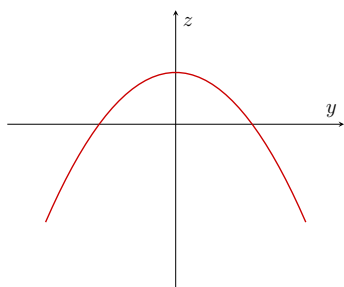


Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. Slicing the surface with vertical planes $x = k$ parallel to the yz -plane yields parabolic trace curves.

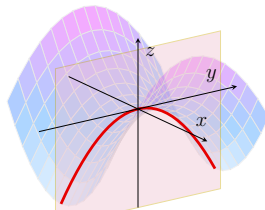
Trace at $x = -1$



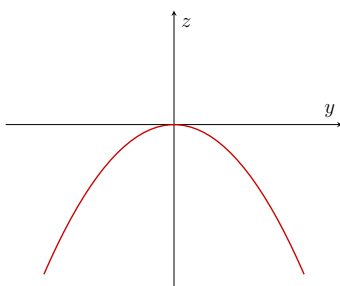
$$z = 1 - y^2$$



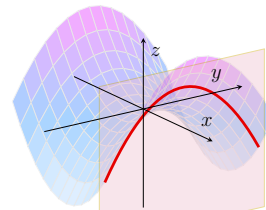
Trace at $x = 0$



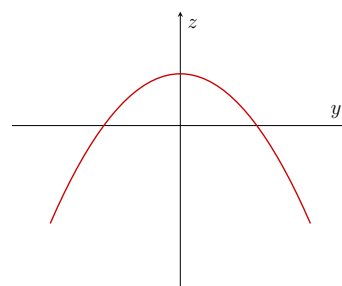
$$z = -y^2$$



Trace at $x = 1$



$$z = 1 - y^2$$



Ellipsoids are another family of quadric surfaces. The general equation of an ellipsoid is given by:

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1$$

5 Planes and Surfaces

5.1 Determining a Plane

A plane $\mathcal{P}$ in $\mathbb{R}^3$ can be uniquely determined by any of the following geometric data:

- A point Q on the plane and a normal vector $\vec{n} \perp \mathcal{P}$.
- Three non-collinear points.
- A general linear equation $ax + by + cz + d = 0$.
- Two intersecting lines (or two non-parallel direction vectors).
- Two parallel distinct lines.
- A line $\mathcal{L}$ and a point $Q \notin \mathcal{L}$.

5.2 Mathematical Formulations of a Plane

Case 1: Plane passing through the origin $O(0, 0, 0)$

A vector $\vec{v} = \langle x, y, z \rangle$ lies in $\mathcal{P} \iff \vec{v} \perp \vec{n}$, where $\vec{n} = \langle a, b, c \rangle$:

$$\vec{v} \in \mathcal{P} \iff \vec{v} \cdot \vec{n} = 0 \iff ax + by + cz = 0$$

In set notation:

$$\mathcal{P} = \{ \langle x, y, z \rangle \mid ax + by + cz = 0 \}$$

Case 2: Plane passing through a point

$Q(x_0, y_0, z_0)$

Let Q be a known point on $\mathcal{P}$ with position vector $\overrightarrow{OQ} = \langle x_0, y_0, z_0 \rangle$. Any point $W(x, y, z)$ on the plane satisfies:

$$(\overrightarrow{OW} - \overrightarrow{OQ}) \perp \vec{n} \iff (\overrightarrow{OW} - \overrightarrow{OQ}) \cdot \vec{n} = 0$$

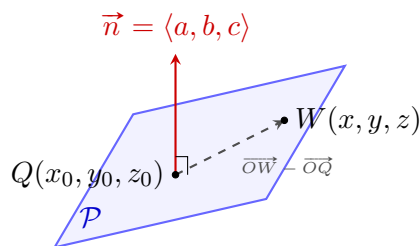
Expanding this dot product yields the point-normal form:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

or equivalently:

$$ax + by + cz = ax_0 + by_0 + cz_0$$

We can always scale the coefficients a, b, c, d in $ax + by + cz = d$ by $\frac{1}{\|\vec{n}\|}$ to obtain a unit normal vector $\hat{n} = \frac{\langle a, b, c \rangle}{\sqrt{a^2 + b^2 + c^2}}$.



5.3 Properties of Normal Vectors

For any set of vectors $\{\vec{u}, \vec{v}, \vec{w}\}$ where $\vec{v} \nparallel \vec{w}$, the following statements are equivalent:

- $\vec{u} \perp \vec{v}$ and $\vec{u} \perp \vec{w}$.
- $\vec{u} \parallel (\vec{v} \times \vec{w})$.
- $\vec{u} \perp \mathcal{P}$, where $\mathcal{P} = \text{Span}\{\vec{v}, \vec{w}\}$.

Problem 5.1: Finding Normal Vectors

Find a normal vector $\vec{n}$ to the plane $\mathcal{P}$ given by the equation $(x - 1) - (y - 1) = 0$.

Solution 5.1

The equation simplifies to $x - y = 0$. We can extract or compute the normal vector in two ways:

Method 1 (Direct Inspection): From the general form $ax + by + cz = d$, the coefficients of x, y, z directly give the normal vector:

$$\vec{n} = \langle 1, -1, 0 \rangle$$

Method 2 (Cross Product of Parallel Plane Vectors): Two vectors parallel to the plane $\mathcal{P}$ are $\vec{v}_1 = \langle 0, 0, 1 \rangle$ and $\vec{v}_2 = \langle 1, 1, 0 \rangle$. Taking their cross product gives:

$$\vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \vec{i}(0 - 1) - \vec{j}(0 - 1) + \vec{k}(0 - 0) = \langle -1, 1, 0 \rangle$$

Note that $\langle -1, 1, 0 \rangle = -\langle 1, -1, 0 \rangle$, which represents the same normal line direction.

Problem 5.2: Finding a Plane From Three Points

Find the equation of the plane containing the points $P(1, 0, 1)$, $Q(0, 2, 0)$, and $R(2, 1, 0)$.

Solution 5.2

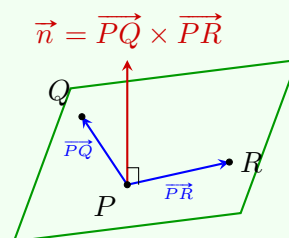
Step 1: Construct two displacement vectors in the plane

$$\overrightarrow{PQ} = \langle 0 - 1, 2 - 0, 0 - 1 \rangle = \langle -1, 2, -1 \rangle$$

$$\overrightarrow{PR} = \langle 2 - 1, 1 - 0, 0 - 1 \rangle = \langle 1, 1, -1 \rangle$$

Step 2: Compute the normal vector via cross product

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 2 & -1 \\ 1 & 1 & -1 \end{vmatrix} = \langle -1, -2, -3 \rangle$$



Step 3: Write the plane equation using point $P(1, 0, 1)$

$$-1(x - 1) - 2(y - 0) - 3(z - 1) = 0$$

Simplifying yields:

$$x + 2y + 3z = 4$$

6 Spheres and Planes

6.1 Determining a Sphere

A sphere in $\mathbb{R}^3$ can be uniquely determined by any of the following sets of data:

- A center point $(a, b, c) \in \mathbb{R}^3$ and a radius $R \in \mathbb{R}^{>0}$.
- Two points representing the endpoints of a diameter.
- Four non-coplanar points.

- A plane tangent to the sphere together with three points lying on the sphere.

6.2 Plane-Sphere Intersections

Example 6.1. Consider the intersection of the sphere $x^2 + y^2 + z^2 = 4$ with the line $\ell(t) = (t, 0, 1)$ passing through $(1, 0, 1)$ and $(-1, 0, 1)$.

Substituting $\ell(t)$ into the sphere equation yields:

$$\begin{aligned} t^2 + 0^2 + 1^2 &= 4 \\ t^2 + 1 &= 4 \\ t^2 = 3 &\implies t = \pm\sqrt{3} \end{aligned}$$

Thus, the line intersects the sphere at $\ell(\sqrt{3}) = (\sqrt{3}, 0, 1)$ and $\ell(-\sqrt{3}) = (-\sqrt{3}, 0, 1)$.

6.3 Traces of Surfaces

Definition 6.1: Traces

A **trace** of a surface is the intersection curve obtained by intersecting the surface with a parallel coordinate plane of the form $x = k$, $y = k$, or $z = k$.

Example 6.2 (Traces of a Sphere). Consider the sphere given by:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

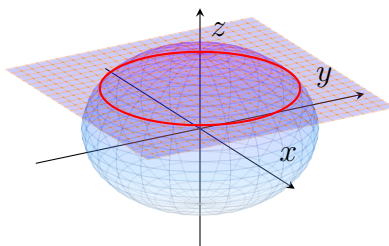
Taking the trace with the plane $z = k$ yields the equation:

$$(x - a)^2 + (y - b)^2 = R^2 - (k - c)^2$$

Depending on the plane distance k , we have three distinct cases:

- **Case 1** ($|k - c| = R$): The trace is a single point at (a, b, k) .
- **Case 2** ($|k - c| < R$): The trace is a circle of radius $\sqrt{R^2 - (k - c)^2}$ centered at (a, b, k) .

- **Case 3** ($|k - c| > R$): The trace is empty (no real intersection).



Example 6.3. Find the trace of the plane $\mathcal{P} = \{(x, y, z) \mid -2x + 6y - 5z = 4\}$ with the plane $x = -1$:

$$-2(-1) + 6y - 5z = 4$$

$$2 + 6y - 5z = 4$$

$$6y - 5z = 2 \implies z = \frac{6}{5}y - \frac{2}{5}$$

The trace is a straight line in the plane $x = -1$.

6.4 Quadric Surfaces

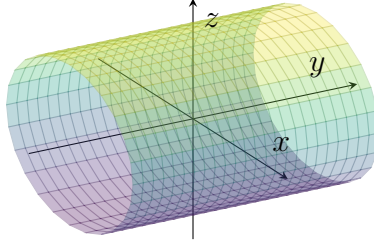
A general **quadric surface** in $\mathbb{R}^3$ is given by the second-degree equation:

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fzx + ax + by + cz + d = 0$$

When $D = E = F = 0$, the surface is in **standard position** (axes aligned with coordinate axes).

Example 6.4. The surface $x^2 + z^2 = 1$ is a circular cylinder aligned along the y -axis with radius 1.

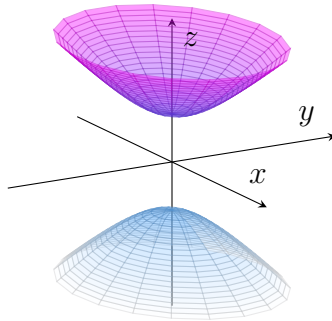
- $z = 0$ trace: $x^2 = 1 \implies x = \pm 1$ (parallel lines along y).
- $x = 0$ trace: $z^2 = 1 \implies z = \pm 1$ (parallel lines along y).



Example 6.5. Consider the surface given by $x^2 + 4y^2 - z^2 = -1$.

Traces:

- $x = 0$ **trace:** $4y^2 - z^2 = -1 \implies z = \pm\sqrt{4y^2 + 1}$. As $|y| \rightarrow \infty$, $z \approx \pm 2y$ (Hyperbola).
- $y = 0$ **trace:** $x^2 - z^2 = -1 \implies z = \pm\sqrt{x^2 + 1}$. As $|x| \rightarrow \infty$, $z \approx \pm x$ (Hyperbola).
- $z = k$ **trace:** $x^2 + 4y^2 = k^2 - 1$:
 - $|k| = 1$: Single point at $(0, 0, \pm 1)$.
 - $|k| < 1$: No trace (empty set).
 - $|k| > 1$: Ellipse given by $x^2 + 4y^2 = k^2 - 1$.



Why do cross-sectional ellipses fall on the surface?

For any fixed height $z = k$ with $|k| > 1$, substituting $z = k$ fixes the value $k^2 - 1 > 0$ on the right side of $x^2 + 4y^2 = k^2 - 1$. This traces a locus of points (x, y, k) whose horizontal distance components continuously satisfy the surface equation for that slice.

7 Vector-Valued Functions

7.1 Coordinate Systems in 3D Space

Specifying a point or giving directions in space can be accomplished in multiple ways:

- Giving a direction and a distance.
- Using Cartesian coordinates $(d_1, d_2, d_3) = (x, y, z)$.
- Specifying an angle relative to a reference axis along with a magnitude.
- Referencing a relative landmark.
- Using geographic coordinates like latitude and longitude.

7.1.1 Polar Coordinates (2D)

In the xy -plane, a point is specified by its radial distance r from the origin and an angle θ relative to the positive x -axis: (r, θ) .

$$r = \sqrt{x^2 + y^2} \quad (r \geq 0)$$

While $\theta = \arctan\left(\frac{y}{x}\right)$ is commonly used, it is undefined when $x = 0$. The foundational algebraic transformation between systems is given by:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

7.1.2 Cylindrical Coordinates

Cylindrical coordinates (r, θ, z) extend polar coordinates into 3D space by preserving the standard Cartesian z -axis:

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

7.1.3 Spherical Coordinates

Spherical coordinates are represented as (ρ, θ, ϕ) , where ρ is the distance from the origin, θ is the azimuthal angle in the xy -plane, and ϕ is the polar angle measured down from the positive z -axis.

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

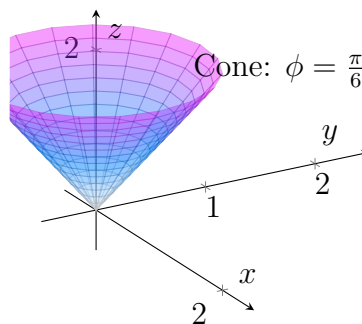
$$z = \rho \cos \phi$$

The standard domain boundaries are:

$$\rho \geq 0$$

$$0 \leq \theta < 2\pi$$

$$0 \leq \phi \leq \pi$$



7.2 Coordinate System Transformations

Problem 7.1: Conversion Between Coordinate Systems

Convert the surface equation $x^2 + \frac{y^2}{a^2} = z^2$ into cylindrical and spherical coordinates.

Solution 7.1

Cylindrical Coordinates:

Substituting $x = r \cos \theta$, $y = r \sin \theta$, and $z = z$ into the original equation yields:

$$\begin{aligned} (r \cos \theta)^2 + \frac{(r \sin \theta)^2}{a^2} &= z^2 \\ r^2 \cos^2 \theta + \frac{r^2 \sin^2 \theta}{a^2} &= z^2 \\ r^2 \left(\cos^2 \theta + \frac{\sin^2 \theta}{a^2} \right) &= z^2 \end{aligned}$$

Spherical Coordinates:

Substituting $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, and $z = \rho \cos \phi$ into the original equation yields:

$$\begin{aligned} (\rho \sin \phi \cos \theta)^2 + \frac{(\rho \sin \phi \sin \theta)^2}{a^2} &= (\rho \cos \phi)^2 \\ \rho^2 \sin^2 \phi \cos^2 \theta + \frac{\rho^2 \sin^2 \phi \sin^2 \theta}{a^2} &= \rho^2 \cos^2 \phi \\ \rho^2 \sin^2 \phi \left(\cos^2 \theta + \frac{\sin^2 \theta}{a^2} \right) &= \rho^2 \cos^2 \phi \end{aligned}$$

Assuming $\rho \neq 0$, we divide both sides by ρ^2 :

$$\begin{aligned}\sin^2 \phi \left(\cos^2 \theta + \frac{\sin^2 \theta}{a^2} \right) &= \cos^2 \phi \\ \cos^2 \theta + \frac{\sin^2 \theta}{a^2} &= \frac{\cos^2 \phi}{\sin^2 \phi} \\ \cos^2 \theta + \frac{\sin^2 \theta}{a^2} &= \cot^2 \phi\end{aligned}$$

7.3 Vector-Valued Functions

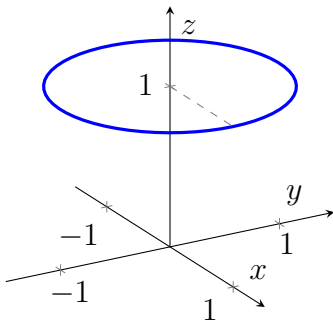
Definition 7.1: Vector-Valued Functions

A vector-valued function maps one or more real parameters to a space vector:

$$\vec{r}(t) = \langle x(t), y(t), z(t) \rangle, \quad \vec{r} : I \rightarrow \mathbb{R}^3$$

where $I \subseteq \mathbb{R}$ is an interval. This traces out a spatial curve parametrized by t .

Example 7.1. Consider the vector function $\vec{r}(t) = \langle \cos t, \sin t, 1 \rangle$ for $0 \leq t < 2\pi$. This describes a unit circle in 3D space elevated to the horizontal plane $z = 1$.



Problem 7.2: Finding Collisions and Intersections

Let $\vec{r}_1(t) = \left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle$ and $\vec{r}_2(t) = \langle t, t^2 \rangle$. Find all collision points and structural intersections between the curves.

Solution 7.2

1. Collision Points (Same location at same time t):

Setting $\vec{r}_1(t) = \vec{r}_2(t)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \langle t, t^2 \rangle$$

Equating the individual components gives the system:

$$\begin{aligned} t^2 + \frac{t}{2} &= t \\ t - t^3 &= t^2 \end{aligned}$$

Solving the first equation for t :

$$\begin{aligned} t^2 + \frac{t}{2} - t &= 0 \\ t^2 - \frac{t}{2} &= 0 \\ t \left(t - \frac{1}{2} \right) &= 0 \implies t = 0 \quad \text{or} \quad t = \frac{1}{2} \end{aligned}$$

Now, we verify both parameters in the second component equation $t - t^3 = t^2$:

$$\begin{aligned} \text{For } t = 0 : \quad 0 - (0)^3 &= (0)^2 \implies 0 = 0 \quad (\text{Valid}) \\ \text{For } t = \frac{1}{2} : \quad \frac{1}{2} - \left(\frac{1}{2} \right)^3 &= \frac{1}{2} - \frac{1}{8} = \frac{3}{8} \\ \left(\frac{1}{2} \right)^2 &= \frac{1}{4} = \frac{2}{8} \neq \frac{3}{8} \quad (\text{Invalid}) \end{aligned}$$

Therefore, $t = 0$ is the only valid collision time. Evaluating $\vec{r}_1(0)$:

$$\vec{r}_1(0) = \left\langle 0^2 + \frac{0}{2}, 0 - 0^3 \right\rangle = \langle 0, 0 \rangle$$

The collision point is $(0, 0)$.

2. Intersection Points (Same location at potentially different times t and

s):

Setting $\vec{r}_1(t) = \vec{r}_2(s)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \langle s, s^2 \rangle$$

Equating components yields:

$$s = t^2 + \frac{t}{2} \tag{1}$$

$$t - t^3 = s^2 \tag{2}$$

Substituting equation (1) into equation (2):

$$\begin{aligned} t - t^3 &= \left(t^2 + \frac{t}{2} \right)^2 \\ t - t^3 &= t^4 + t^3 + \frac{t^2}{4} \\ 0 &= t^4 + 2t^3 + \frac{t^2}{4} - t \\ 0 &= t \left(t^3 + 2t^2 + \frac{t}{4} - 1 \right) \end{aligned}$$

Setting each factor to zero:

$$t = 0 \implies s = 0^2 + \frac{0}{2} = 0$$

$\implies$ Intersection point $(0, 0)$

$$t^3 + 2t^2 + \frac{t}{4} - 1 = 0 \implies t \approx 0.576$$

Solving for s when $t \approx 0.576$:

$$s = (0.576)^2 + \frac{0.576}{2}$$

$$s \approx 0.332 + 0.288 \approx 0.620$$

Evaluating the spatial coordinates using $\vec{r}_2(s)$:

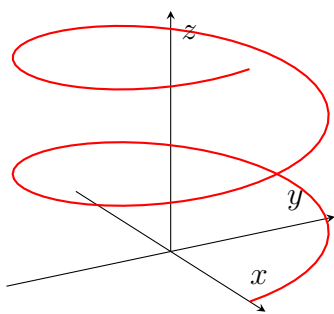
$$x = s \approx 0.620$$

$$y = s^2 \approx (0.620)^2 \approx 0.384$$

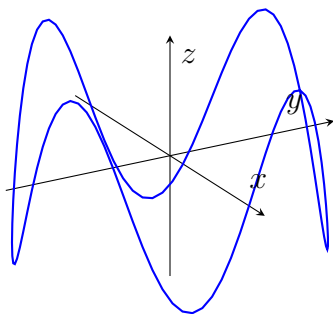
Hence, the two spatial intersection points are $(0, 0)$ and $(0.620, 0.384)$.

7.4 Gallery of Parametric Space Curves

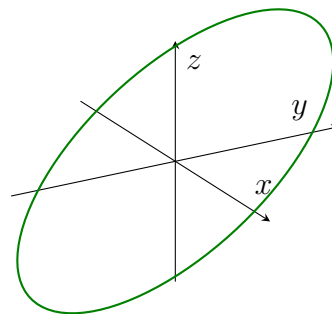
Below are standard parametric 3D space curves used in multivariable trajectory analysis:



$$\vec{r}(t) = \langle \cos t, \sin t, t \rangle$$



$$\vec{r}(s) = \langle \cos s, \sin s, \sin(4s) \rangle$$



$$\vec{r}(t) = \langle \cos t, \sin t, 4 \sin t \rangle$$